Stage 2: Expanded Finite-Difference, CN, LCP, and PSOR Notes

Free-Boundary-Aware Risk Surfaces for American Options

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Abstract

This teaching-oriented report derives the finite-difference operator, tridiagonal matrix, Crank–Nicolson matrices, boundary adjustments, American linear complementarity problem, and projected SOR iteration. All numerical statements retain the scope of the concise Stage 2 validation report.

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1 Stage 2: From the PDE to CN/PSOR

1.1 Purpose

This stage explains the finite-difference matrix construction in enough detail to connect every mathematical coefficient with the arrays used by the solver. The target is a reader who understands derivatives and matrix multiplication but has not previously implemented an American-option LCP.

1.2 Method

1.2.1 Spot and time grids

Truncate the unbounded spot domain to $[0, S_{\max}]$. Choose M equal spot intervals and N equal time intervals:

$$S_i = i\Delta S, \qquad i = 0, \ldots, M, \qquad \Delta S = \frac{S_{\max}}{M},$$

$$\tau_n = n\Delta\tau, \qquad n = 0, \ldots, N, \qquad \Delta\tau = \frac{T}{N}.$$

Write U_i^n for the approximation to $U(S_i, \tau_n)$. At a fixed time, the interior unknown vector is

$$U^n = (U_1^n, U_2^n, \ldots, U_{M-1}^n)^\top.$$

> **Common confusion.** The index i moves across spot and the index n moves across time to maturity. U_i^n is one scalar. U^n is the column vector containing all interior spot values at time level n.

> **Common confusion.** U_0^n and U_M^n are not missing unknowns. They are values at the two domain boundaries $S = 0$ and $S = S_{\max}$, where an option-specific boundary condition supplies the value.

1.3 Mathematical Formulation

Step-by-step derivation: central differences

At an interior node, central differences give

$$U_S(S_i, \tau_n) \approx \frac{U_{i+1}^n - U_{i-1}^n}{2\Delta S},$$

$$U_{SS}(S_i, \tau_n) \approx \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{(\Delta S)^2}.$$

Substitute them into

$$\mathcal{L}U = \frac{1}{2}\sigma^2 S^2 U_{SS} + (r - q)SU_S - rU.$$

At S_i,

$$(L_hU)_i = \frac{1}{2}\sigma^2 S_i^2 \frac{U_{i+1} - 2U_i + U_{i-1}}{(\Delta S)^2} + (r-q)S_i \frac{U_{i+1} - U_{i-1}}{2\Delta S} - rU_i.$$

Collect the coefficients of the three neighboring values:

$$(L_hU)_i = a_i U_{i-1} + b_i U_i + c_i U_{i+1},$$

where

$$a_i = \frac{\frac{1}{2}\sigma^2 S_i^2}{(\Delta S)^2} - \frac{(r-q)S_i}{2\Delta S},$$

$$b_i = -\frac{\sigma^2 S_i^2}{(\Delta S)^2} - r,$$

$$c_i = \frac{\frac{1}{2}\sigma^2 S_i^2}{(\Delta S)^2} + \frac{(r-q)S_i}{2\Delta S}.$$

1.3.1 Where the coefficients come from

Source	Contribution to a_i	Contribution to b_i	Contribution to c_i
Diffusion	$+\frac{1}{2}\sigma^2 S_i^2/(\Delta S)^2$	$-\sigma^2 S_i^2/(\Delta S)^2$	$+\frac{1}{2}\sigma^2 S_i^2/(\Delta S)^2$
Drift	$-(r-q)S_i/(2\Delta S)$	0	$+(r-q)S_i/(2\Delta S)$
Discount	0	$-r$	0

> **Reader intuition.** Diffusion spreads weight symmetrically to the left and right. Drift shifts the two off-diagonal weights in opposite directions. Discounting affects only the current node and therefore appears only in the diagonal coefficient.

Step-by-step derivation: a small matrix with $M = 4$

With nodes S_0, S_1, S_2, S_3, S_4, there are three interior unknowns:

$$U^n = (U_1^n, U_2^n, U_3^n)^\top.$$

The interior matrix is

$$L_h = \begin{bmatrix} b_1 & c_1 & 0 \\ a_2 & b_2 & c_2 \\ 0 & a_3 & b_3 \end{bmatrix}.$$

The full stencil equations are

$$(L_hU)_1 = a_1 U_0 + b_1 U_1 + c_1 U_2,$$
$$(L_hU)_2 = a_2 U_1 + b_2 U_2 + c_2 U_3,$$
$$(L_hU)_3 = a_3 U_2 + b_3 U_3 + c_3 U_4.$$

Matrix multiplication by the displayed 3×3 matrix contains only the interior terms. The omitted $a_1 U_0$ and $c_3 U_4$ are known boundary contributions.

Common confusion. The first matrix row has no column for U_0, and the last row has no column for U_4, because the matrix maps the three-component interior vector to another three-component vector. Only the first stencil touches the lower boundary; only the last stencil touches the upper boundary.

Step-by-step derivation: Crank–Nicolson time stepping

Crank–Nicolson averages the spatial operator at the old and new time levels:

$$\frac{U^{n+1} - U^n}{\Delta\tau} = \frac{1}{2}L_h U^{n+1} + \frac{1}{2}L_h U^n.$$

Multiply by $\Delta\tau$:

$$U^{n+1} - U^n = \frac{\Delta\tau}{2}L_h U^{n+1} + \frac{\Delta\tau}{2}L_h U^n.$$

Move the new-time operator to the left:

$$U^{n+1} - \frac{\Delta\tau}{2}L_h U^{n+1} = U^n + \frac{\Delta\tau}{2}L_h U^n.$$

Define

$$A = I - \frac{\Delta\tau}{2}L_h, \qquad B = I + \frac{\Delta\tau}{2}L_h.$$

Then

$$AU^{n+1} = BU^n + g^{n,n+1},$$

where $g^{n,n+1}$ collects known boundary values.

Common confusion. A multiplies U^{n+1} because its terms were gathered on the left. B multiplies U^n because its terms remain on the right. The signs are a consequence of rearrangement, not an arbitrary naming convention.

1.3.2 What one row means

Ignoring the boundary adjustment momentarily, an interior row i reads

$$\begin{aligned} -\frac{\Delta\tau}{2}a_i U_{i-1}^{n+1} + \left(1 - \frac{\Delta\tau}{2}b_i\right)U_i^{n+1} - \frac{\Delta\tau}{2}c_i U_{i+1}^{n+1} \\ = \frac{\Delta\tau}{2}a_i U_{i-1}^{n} + \left(1 + \frac{\Delta\tau}{2}b_i\right)U_i^{n} + \frac{\Delta\tau}{2}c_i U_{i+1}^{n}. \end{aligned}$$

This is the scalar equation represented by row i of A and B.

Step-by-step derivation: why boundary values appear on the right-hand side

For the first interior row, the lower-boundary terms at the two time levels are

$$a_1 U_0^{n+1}, \qquad a_1 U_0^n.$$

Moving the new-time boundary term to the right gives the combined contribution

$$\frac{\Delta\tau}{2} a_1 \left(U_0^n + U_0^{n+1}\right).$$

Similarly, the last row receives

$$c_{M-1} U_M^{n+1}, \qquad c_{M-1} U_M^n,$$

which combine as

$$\frac{\Delta\tau}{2} c_{M-1} \left(U_M^n + U_M^{n+1}\right).$$

Hence

$$g^{n,n+1} = \frac{\Delta\tau}{2} \begin{bmatrix} a_1(U_0^n + U_0^{n+1}) \\ 0 \\ \vdots \\ 0 \\ c_{M-1}(U_M^n + U_M^{n+1}) \end{bmatrix}.$$

> **Implementation connection.** This matches `src/solvers/cn_psor.py`: the first right-hand-side entry is increased by $\frac{1}{2}\Delta\tau\, a_1(U_0^n + U_0^{n+1})$, and the last by $\frac{1}{2}\Delta\tau\, c_{M-1}(U_M^n + U_M^{n+1})$.

1.3.3 Boundary conditions used in the project

For an American put,

$$U(0,\tau) = K, \qquad U(S_{\max},\tau) = 0.$$

At zero spot, immediate exercise returns K. At a sufficiently large spot, the put value approaches zero.

For a dividend-paying American call,

$$U(0,\tau) = 0,$$

$$U(S_{\max},\tau) = \max\left(S_{\max} - K,\; S_{\max} e^{-q\tau} - K e^{-r\tau},\; 0\right).$$

The upper boundary retains both immediate-exercise value and the discounted European far-field expression.

For European validation with projection disabled, the put uses

$$U(0,\tau) = K e^{-r\tau}, \qquad U(S_{\max},\tau) = 0,$$

and the call uses

$$U(0,\tau) = 0, \qquad U(S_{\max},\tau) = S_{\max} e^{-q\tau} - K e^{-r\tau}.$$

Step-by-step derivation: the American linear complementarity problem

Let

$$d^n = BU^n + g^{n,n+1}.$$

A European step solves the equality $AU^{n+1} = d^n$. For an American option, that equality can produce a value below the payoff. The discrete obstacle problem is instead

$$AU^{n+1} \geq d^n, \qquad U^{n+1} \geq \Phi,$$

$$\left(U^{n+1} - \Phi\right)^\top \left(AU^{n+1} - d^n\right) = 0.$$

The first inequality requires a nonnegative discrete PDE residual. The second enforces the payoff obstacle. The final equality imposes complementarity: each node is either continuing with zero PDE residual or exercising with zero premium.

> **Common confusion.** An American option is not priced by solving the European system and then reporting whichever nodes happen to lie above the payoff. The obstacle is part of the system itself, so the solution method must enforce it during the solve.

Step-by-step derivation: projected successive over-relaxation

At one time step, write $Ax = d$ for the interior system and let Φ_i be the obstacle at node i. Starting from an old iterate $x^{(k)}$, a Gauss–Seidel candidate is

$$y_i^{(k+1)} = \frac{d_i - \sum_{j<i} A_{ij} x_j^{(k+1)} - \sum_{j>i} A_{ij} x_j^{(k)}}{A_{ii}}.$$

The first sum uses already updated entries; the second uses entries from the previous sweep. Relaxation gives

$$x_i^{\text{relaxed}} = x_i^{(k)} + \omega \left(y_i^{(k+1)} - x_i^{(k)}\right).$$

Projection then enforces the obstacle:

$$x_i^{\text{new}} = \max\left(\Phi_i, x_i^{\text{relaxed}}\right).$$

The parameter $\omega \in (0, 2)$ controls relaxation. Values above one may accelerate convergence for suitable systems, but excessive relaxation can destabilize the iteration.

1.3.4 Stopping and complementarity diagnostics

The implemented iteration stops when

$$\|x^{(k+1)} - x^{(k)}\|_\infty < \text{tol},$$

or reports failure after the maximum number of sweeps. A small update is necessary but is not the full financial diagnostic. After solving, define

$$z = Ax - d.$$

The LCP checks are

$$x - \Phi \geq 0, \qquad z \geq 0, \qquad (x - \Phi)^\top z \approx 0.$$

These separately test obstacle feasibility, residual feasibility, and complementarity.

> **Reader intuition.** PSOR alternates two actions: move toward the linear-system solution, then clip the result back to the financially admissible side of the payoff. Convergence is credible only when both the numerical update and the LCP residual checks are small.

1.4 Implementation Details

The tridiagonal operator is exposed by `src/solvers/cn_psor.py` for sign and matrix tests. Boundary formulas live in `src/solvers/boundary_conditions.py`. The code uses the payoff as the initial condition at $\tau = 0$, forms A and B once per solve, adjusts the first and last right-hand-side entries at every time step, and applies PSOR only when American projection is enabled.

1.5 Validation and Diagnostics

The existing solver audit tests verify the operator on a linear function, the signs in A and B, put and dividend-call boundary values, and PSOR projection/complementarity. European closed-form and no-dividend call controls provide independent checks beyond internal matrix consistency.

1.6 Results

The verified smoke-grid maximum error is 2.782×10^{-4} for both the European put validation and the no-dividend American call control. The American put obstacle violation is reported as zero, with complementarity error 4.009×10^{-12}. These are engineering-scale validation results, not a claim of universal or publication-grade accuracy.

1.7 Failure Modes and Limitations

Uniform grids resolve the payoff kink and free boundary only approximately. Finite-domain truncation can contaminate the interior if $S_{\max}$ is too small. Crank–Nicolson begins from a nonsmooth payoff, and Rannacher smoothing is not active in the current solver. These limitations are particularly important for Gamma near maturity.

1.8 Conclusion and Next Step

The PDE has become a boundary-adjusted LCP at each time step. Later stages extract the exercise boundary and Greeks, study resolution and runtime, and use only diagnostic-passing solver outputs for machine-learning labels.